International Biodeterioration & Biodegradation 195 (2024) 105887

Contents lists available at ScienceDirect

International Biodeterioration & Biodegradation

journal homepage: www.elsevier.com/locate/ibiod

Bio-based plastics, biodegradable plastics, and compostable plastics: biodegradation mechanism, biodegradability standards and environmental stratagem

Sabzoi Nizamuddin [a,b,*], Abdul Jabbar Baloch [c], Chengrong Chen [b,**], Muhammad Arif [d], Nabisab Mujawar Mubarak [e,f]

[a] Water Regulation Division, Grampians Wimmera Mallee Water (GWMWater) corporation, Horsham, Victoria, 3400, Australia
[b] Australian Rivers Institute and School of Environment and Science, Griffith University, Nathan Campus, 4111, Queensland, Australia
[c] Department of Civil Engineering, NED University of Engineering and Technology, Karachi Sindh, Pakistan
[d] Department of Mechanical and Manufacturing Engineering, Pak-Austria Fachhochschule Institute of Applied Sciences and Technology, Haripur, Khyber Pakhtunkhwa, Pakistan
[e] Petroleum and Chemical Engineering, Faculty of Engineering, Universiti Teknologi Brunei, Bandar Seri Begawan, BE1410, Brunei Darussalam
[f] Department of Biosciences, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, India

A R T I C L E I N F O

Keywords:
Biodegradability
Compostability
Bioplastic materials
Environmental issues
International standards

A B S T R A C T

Conventional polymers are environmentally damaging materials; therefore, global efforts are being made to gradually replace these conventional polymers with bio-based, biodegradable, and compostable plastics due to claims of being more sustainable than petroleum-based plastics. However, such claims may not be based on reality, and unregulated bio plastics may cause environmental anarchy similar to conventional plastics. The degradation of bioplastics has received significant attention because it is the parameter used to evaluate their end-of-life disposal and to assess their environmental shortcomings - where the bioplastics which degrade completely in different environments, thus, considered as an environmental-friendly polymers. Upon disposal, the bioplastics decompose in a bio-active medium by microorganisms such as algae, bacteria, and fungi or to humus, water, and CO_2 by marine water. Different standardization and certification bodies have set the standards for bioplastics, compostable, and biodegradable plastics to evaluate the environmental constraints of bioplastics. These standards support various industries in creating bioplastics. Thus, it is important to harness the regulatory power to bring all the standardization and certification bodies (both at the national and international levels) together in setting standards with a high threshold to classify bio-based plastics, biodegradable plastics, and compostable plastics.

1. Introduction

Plastics are defined as a subset of polymers containing long-chain organic molecules with desired properties (Wang et al., 2022), having various advantages compared to wood, ceramics, and metals. For instance, plastics are light in weight, cheaper, chemically inert, adaptable, pliable, easily shaped, and multi-functional (Gu, 2017). Owing to the above-mentioned characteristics, the global production of plastics has increased significantly. The worldwide production of plastics was anticipated to have an annual growth of 8.5% from 1950 to 2015 and global plastic production is expected to surpass 800 million tons before 2050 (Sardon et al., 2018). A variety of plastic materials are available commercially. Among them, seven plastics are more dominant, accounting for 70–90% of total plastic demand. They are polyethylene, low density polyethylene, high density polyethylene, polyvinyl chloride, polypropylene, polyethylene terephthalate, and polystyrene (Storz, 2013).

Although plastic production is in large quantities with a wide range of applications, they typically have a short service life. At the end of their service lives, plastics originally high-value materials, have no economic value as waste material due their end-of-life disposal/use methods being unknown and only a very small part (<9%) of these waste plastics is

* Corresponding author. Water Regulation Division Grampians Wimmera Mallee Water (GWMWater) corporation, Horsham 3400, Victoria, Australia.
** Corresponding author.
E-mail addresses: nizamuddin248@gmail.com (S. Nizamuddin), c.chen@griffith.edu.au (C. Chen).

https://doi.org/10.1016/j.ibiod.2024.105887
Received 13 April 2024; Received in revised form 30 July 2024; Accepted 15 August 2024
Available online 20 August 2024

recycled (Stubbins et al., 2021). Over 80% of plastics become useless and then enter the environment (rivers, lands, and oceans), leading to a heavy economic loss, around 80–100 billion USD per year, togther with causing serious environmental concerns (Garcia et al., 2017; Ragaert et al., 2017; MacLeod et al., 2021). Plastics undergo a slow degradation process and stay in bulk form for hundreds of years due to high molecular weight covalent bonds forming a static network causing undesirable degradation profiles (Webber et al., 2022). Furthermore, waste plastics disposed off in water or land degrade into micro or nano-particles (Guan et al., 2021). These small particles of plastics contaminate aquatic lives through ingestion and entanglement, transfer toxins to food chains and finally affect human beings (Seltenrich, 2016; Law, 2017; Rahimi et al., 2017). To mitigate and/or minimize the environmental effects related to plastics disposal and end-of-life concerns of these plastics, it is suggested to develop materials with biodegradability and compostability, which can degrade faster (Christensen et al., 2019) and can be utilized as a substitute for traditional polymers (Hillmyer, 2017; Song et al., 2009).

The “biodegradability” and “composability” of bioplastics have received significant attention among industry partners, researchers, and academicians due to the recalcitrant nature of plastics, the large amounts of plastics disposed of every year, and the accumulation of plastics in the ecosystem (Gu, 2021). However, there is confusion on the definition of biodegradable plastics, bio-based plastics, and compostable plastics – these terms are often used interchangeably, but they are not synonymous. Companies need to understand the distinctions between each category to accurately and honestly market their products. Also, consumers need to understand these terms to make educated purchasing decisions and properly dispose of plastic products at the end of use (Nazareth et al., 2022). Therefore, it is important to design and formulate the standards for each bioplastic types i.e., bio-based, biodegradable, and compostable plastics to differentiate between them.

Several organizations, including the American Certification System of Biodegradable Products Institute (BPI), the US Composting Council (USCC), the American Society for Testing and Materials (ASTM), the Environment & Plastics Industry Council, British Plastics Federation, Japan’s GreenPla program, and European Committee for Standardization (CEN), are involved in setting the standards for compostable, bio-based, and biodegradable plastics. These standards support industries in creating bioplastics to meet the increasing demands of environmental-friendly plastics (Narayn et al., 1999).

This review article explains biodegradable plastics, bio-based plastics, and compostable plastics and their derivatives. The standards (describing each bioplastic type) are analyzed, the shortcomings are highlighted, and suggestions are provided to meet the set standards. Moreover, the environmental concerns associated with using conventional plastics are highlighted, and their potential replacement is recommended in the form of bioplastics. A critical overview of the topic is provided, together with significant conclusions, and suggestions are offered to meet the standards of bioplastics in terms of biodegradability and compostability.

2. Bioplastics

2.1. Biodegradable plastics

Biodegradable plastics are defined as plastics that degrade by the actions of natural microorganisms such as algae, bacteria, and fungi (Greene, 2007; Goel et al., 2021). There are different categories of biodegradable plastics originating from petroleum sources such as aliphatic polyesters and co-polyesters, e.g., poly(butylene succinate) – PBS, poly(ε-caprolactone) – PCL, polyesteramides – PEA, Poly(vinyl alcohol) – PVA and their blends. At the commercial scale, PBS is available in the form of aliphatic polyester and possesses tremendous properties such as chemical and thermal resistance, biodegradability, and melt processability; therefore, it can be easily processed in the field of monofilament, textiles into melt blow, flat, multifilament, and split yarn as well as utilized in the process of plastics injected molded products. Thus, it has multiple applications, including packaging film, cutlery, bags, mulch film, containers, and “flushable” hygienic-based products. PCL is easily processed due to the low viscosity of melt and low melting point (60^{o}C). PCL shows strong resistance to oil, water, and chlorine. The flexibility in the PCL chain displays strong elongation at breakage and low modulus. The characteristic applications of PCL include polymer processing, synthetic wound dressings, non-woven fabrics, adhesives, biodegradable films and bottles, orthopedic casts, fertilizers, and pesticide-controlled release processes. PEA is a new variety of thermo-plastics with higher performance and greater biodegradability. BAK 1095 is a good example of poly (estermaide) due to a reduction in the glass and melting temperatures when the amide to ester ratio is increased. BAK 1095 resin is used in the agricultural, food, and horticultural industries. Some useful examples include plant pots, cemetery decorations, agricultural biofilms, one-way dishes, plant clips, and biowaste bags. Furthermore, PVA is a polymer with water-soluble characteristics that relies on fossil fuel sources, including lustre, chemical resistance, good transparency, toughness, and anti-electrostatic properties. It also has strong printability and gas barrier properties. PVA is broadly utilized in film, fabric, textile sizing, plywood manufacturing, emulsifier, thermosetting resins modifier, and pressure-sensitive adhesives. Significantly, PVA has applications as a sizing agent as well as to stabilize dispersion systems. Major applications of PVA include adhesives & coatings due to excellent film-forming capacity, water-soluble films due to high water solubility, textiles & paper because of its flexibility and high tensile strength, ceramic and cement industry to improve the mechanical properties of different ceramics and enhancing the workability and water retention properties of cement.

Bio-based and biodegradable plastics can potentially substitute the most conventional plastics frequently used today. Shen et al. (2009) studied bio-based plastics’ technical substitution potential (bio-based and biodegradable or bio-based and non-biodegradable) for their petrochemical counterparts. It was found that bio-based and biodegradable plastics can replace almost 94% of conventional polymers: 63% substitution by bio-based and non-biodegradable plastics and 31% substitution from bio-based and biodegradable plastics. Similarly, Yano et al. (2014) investigated the substitution potential of containers and packaging in Japan by polylactic acid (PLA) and PLA derivatives. It was found that 86.9% of containers and packaging plastics can be replaced by PLA and PLA derivatives.

2.2. Bio-based plastics

Bio-based plastics are regarded as the largest type of plastics that are resourced from biomass as they originate bio-based, are metabolised in the organic biomass, and their end product is bio-degradable or coupled bio-based and biodegradable. According to the European Bioplastics Organization, bio-based plastic is a plastic material that is either biodegradable, bio-based or features both characteristics (Goel et al., 2021). Based on this definition, bio-based plastics can be classified into three categories: (i) bio-based bioplastics, (ii) biodegradable bioplastics, and (iii) bio-based and biodegradable bioplastics. The term “bio-based” is described as a material or product which is obtained (either partially or fully) from biomass (plants) such as cellulose, sugarcane or corn, etc. (Tokiwa et al., 2009). For instance, polyethylene is produced from ethylene which is produced from alcohol, and this alcohol is produced from biomass. Such polyethylene will be termed bio-polyethylene or bio-based polyethylene (Goel et al., 2021). The bio-based but non-biodegradable plastics are bio-polyethylene (Bio-PE) and polysaccharide derivatives mostly used for packaging applications, polyol-polyurethane used in tyres synthesis, and bio poly (ethylene terephthalate) bio-PET. Examples of bio-based and biodegradable plastics are PLA, PHA, and poly (amino acid) used for medical applications.

The most common bio-based plastics (or biopolymers) are polysaccharides including cellulose and starch derivatives, alginates, and chitosan; proteins including gluten, whey, and casein; lipids including free fatty acids, and bees and carnauba wax; poly hydroxybutyrates (PHB); polycaprolactone (PCL); polyglycolic acid (PGA); PLA; PHA; polyvinyl alcohol (PHA); polybutylene succinate and their blends (Youssef et al., 2018). Among bio-polymer matrices which are used to form bio-based plastics, PLA and starch are on top due to their advantages such as lower cost, ample availability, bio-degradability, bio-compatibility, non-toxicity, and renewability (Takkalkar, 2019). The major types of bio-based plastics are described below in detail.

Among existing bio-based polymers, PLA is the only material produced in large quantities (more than 140,000 tons per year) (Shen et al., 2010), and it requires 25–55% less energy as compared to conventional polymers (Xiao et al., 2012). Hence, it is considered cost-effective and eco-friendly and an ideal alternative of petroleum-driven polymers (Auras et al., 2004, 2011). PLA is regarded as a bio-compatible thermos-plastic polyester formed through fermentation of biomass resources such as starch. Compared with conventional plastics, the key benefit of PLA is that it is easily biodegraded and is produced from agriculture-based biomass renewable sources (Oksman et al., 2006). The chemistry of PLA shows that it is a linear aliphatic thermos-plastic polyester formed from lactic acid fermentation. Lactic acid (2-hydroxy propionic acid) is a chiral molecule and is present in two stereoisomeric forms: L- and D-lactic acid. Lactic acid is supplied from renewable bio-resources including maize, wheat, sugar, starch, sugar beets, and agricultural waste. PLA polymers catalogued from glassy biopolymers with 50–60^{o}C glass temperature to the semicrystalline forms with 130-180 °C melting point, depending upon enantiomeric repeating units (L and D) sequence in the polymer backbone. PLA shows outstanding thermal plasticity, biocompatibility, and mechanical properties. Standardized PLA has strong strength and modulus but is deficient in toughness properties. However, toughness of PLA can be enhanced by copolymerization, blending, and orientation techniques. PLA and its derivatives are useful in various applications such as paper coating, films, fibers, and food packaging. It shows a strong film with excellent barrier characteristics but a low heat seal for packaging process. For fiber, it means that apparel can have outstanding characteristics and better management of moisture, and industrial clothing has good resistance to UV, lower flammability, and better resistance to stains and soil. A blending approach is highly recommended to enhance the performance of PLA and improve numerous properties (Nampoothiri et al., 2010).

PHA is a polyester of numerous hydroxy alkanoates formed by gram-positive and gram-negative bacteria from 75 bacterial strains. PHAs are formed from a broad range of bioproducts resources including fossil resources (mineral, lignite, oil, methane, hard coal), renewable resources (starch, triacylglycerols, cellulose, sucrose), chemicals (4-hydroxybutyric acid, propionic acid), carbon dioxide (CO_2), and by-products (whey, molasses, glycerol). The PHA family possesses outstanding mechanical characteristics ranging from crystal to elastic, which largely rely on the configuration of monomeric units. The functionalized PHAs are synthesized from larger monomers e.g., medium-chain mc1-PHA, attached to C6–C14 monomeric units, which are stereotypically elastomers and are sticky materials that are modifiable to form rubber. At first, PHA and its derivatives were utilized in packaging film industry, primarily in paper coating, containers and bag formation. Comparable to conventionally-based plastics that contain disposable products include nappies, cosmetic containers, shampoo, razors, hygienic materials, utensils, cups, and bottles.

TPS, the polysaccharide of rice, tubers, legumes, lentils, cereals, and wheat, is a broadly existing renewable raw material and the end-product of photosynthetic reaction. In nature, starch is the second most abundant biomass (Le Corre et al., 2010), consisting of two substances, a high-branch polysaccharide-amylopectin and a linear polysaccharide-amylose (García et al., 2015; Rajisha et al., 2014). The structure of amylose consists of linear α-1, 4 glycosidic linkages (Tikapunya et al., 2017). Starch contain more amylose and has higher chain-to-chain interactions, which triggers the production of a highly-entangled network that will not approve the dispersion process (Takkalkar, 2019). On the contrary, branched amylopectin comprises α-1, 4 chains interconnected by α-1, 6 bonds comprising 10–60 glucose elements, and side chains comprising 15–45 glucose units (García et al., 2015). In amylopectin starch, the outer chains create a weak-entangled network, which favors the dispersion within starch-based polymers (Le Corre et al., 2010). Thermoplastic starch is synthesized from natural starch through a plasticizing or swelling agent by employing starch in the compound extruder in the absence of water. The de-structured starch is mainly formed when starch containing more than 5% water is pasted or plasticized under temperature and pressure. Natural starch is non-plasticized in nature because of inter- and intramolecular hydrogen bonds between hydroxyl and starch molecules. The physical characteristics of thermo-plastic starch are strongly prejudiced by the presence of plasticizers, containing thermal, water-absorption, and mechanical properties. At the commercial scale, the first product synthesized from injection-molded thermoplastic starch was the drug delivery capsule Capill, and later products were progressively produced, for example plates, cutlery, food containers, and golf tees. Additionally, extrusion is utilized to make rigid foams, which are useful for loose-fill packaging process.

The chitin morphology strongly resembles a cellulose structure, with hydroxyl groups at the C-2 position of D-glucopyranose residues exchanged by N-acetylamino groups. Chitin signifies the second most abundantly available polysaccharide after occurrence of cellulose in nature. In arthropod shells, the exoskeletons comprise chitin consisting of 20–50% chitin by dry weight. Seafood wastes are utilized and recycled in industry, and after processing, they sourced for the production of chitin. Naturally, chitin is polysaccharide and hydrophobic and is insoluble in water and other organic solvents. The structure of chitin is nonelastic, hard, and of nitrogenous polysaccharide.

Chitosan is an example of a basic polysaccharide. Chitosan is highly-soluble in dilute acids e.g., formic and acetic acids and also has various beneficial features including biodegradability, biocompatibility, antibacterial properties, and hydrophilicity. Before melting, chitin and chitosan - the derivatives of polysaccharides - are strongly degraded due to wide-ranging hydrogen bonding. Chitosan has potential applications in various fields such as functional membranes, biomedicine, flocculation, and wastewater treatment; more specifically has outstanding applications in the field of genetic cargo, tissue engineering, cell culture, and gene delivery approaches. Moreover, chitosan is identified as a strong adsorbent for removing heavy-metal ions and is utilized to shield edible coatings. Additionally, chitosan films also help to improve the shelf-life of vegetables.

2.3. Compostable plastics

Naturally, composting is a decay process in the soil consisting of raw materials such as grass clippings, food wastes, leaves, manure, and biosolids. In composting process, organic debris is converted to stable humic substance which is known as compost. The compost has applications in horticulture, agriculture, and the silviculture industries. Composting is a promising technology for waste management of degradable plastics, especially for those plastics which cannot be recycled due to contamination of organic substances (Ingrao et al., 2017). According to the ASTM D6400 standard, compostable plastic is a material which degrades due to biological procedures occurring in composting and yields water, CO_2, inorganic substances, biomass, and does not leave any visible, distinguishable, or toxic residues (Greene, 2007). Several composting methods include drum reactors, row composting, table composting and tunnel composting. However, it is common for all methods to have similar flow of materials and similar end products (Dilkes-Hoffman et al., 2019a). In the 1980s, compostable polymers were first revolutionized on a commercial scale. The first-generation compostable products were produced from petroleum-based polymers, specifically polyolefins such as polyethylene, combined with starch or organic substances. Through biological process, the compostable plastics undergo a degradation phenomenon to produce water, inorganic components, CO_2, and biomass in a steady process with identified compostable products.

The important characteristics of a substance to be known as "compostable" are mineralizations i.e., bio-degradation into water, biomass, and CO_2, disintegration to compost, and complete bio-degradation during end-use of compost, and achieve required criteria i.e. non-ecotoxicity. Standardized test methods should be carried out to satisfy the requirements. Compostable materials for packaging applications were introduced for reducing the demand of packaging materials coming from conventional sources which were simultaneously retrieved by municipal waste collection systems. An example of such activity was recycling of organic wastes for compostable cutlery were used at the Olympic Games held in Sydney (Australia) in the year 2000. The project was successfully completed where 76% solid waste was recovered by collecting and composting the waste created by nine million visitors.

Compostable plastics can be divided into different categories based on source of origin, i.e., either derived from petrochemical or renewable sources. Compostable plastics from petroleum sources are PBS, PCL, PEA, and PVA. On the other hand, compostable plastics produced from renewable sources are PLA, PHA, TPS, cellulose, chitin, proteins, and their blends. The best approach implemented in formation of compostable polymeric materials is blending of biodegradable polymers. In polymeric science, blending is a common method to enhance unacceptable physical properties and to reduce costs. The foremost blended compostable materials are starch-based polymers that associate low cost starch with high cost plastics due to strong physical properties. For example, Mater-Bi manufactured by Novamont, is formed by blending bio-degradable bioplastics and starch with water or a plasticizer in an extruder. There are three main types of Mater-Bi that are commercially available and used. They include Class Z – which contains polycaprolactone and TPS, Class Y – containing cellulose/cellulose derivatives with TPS, and Class V – consisting of more than 85% TPS.

To attain compostable polymeric materials, better combinations between processing and mechanical properties, composability, cost, and combinations of compostable polymer have been achieved. The main purpose of blending is to overcome drawbacks of low resilience, high sensitivity to moisture, and high shrinkage of starch and also to improve the relevant properties of starch/polymer-based plastics, including mechanical, thermal, and compostable properties. In general, the major applications of polymer/starch compostable plastics include.

- ➢ Packaging: packaging films, technical films, straws, shopping bags, packaging films, technical films, tableware, straws, trays, compost bags, and wrap film
- ➢ Agriculture industry: encapsulating, plastering, slower releasing of active agents like agrochemicals, planting pots, and mulch film
- ➢ Transportation: fillers in tyres
- ➢ Miscellaneous: soluble loose filers, mantling for nets and candies, soluble cotton swabs, cutlery, cups, golf tees, edge protectors, and nappy back sheets

3. Biodegradation of bioplastics and its mechanism

Although a significant number of studies have reported the biodegradability of bioplastics; however, the most of studies have missed or not focused on the chemistry and composition of plastics, methods of enrichment of microorganisms, metabolism of microorganisms & biochemical process involved and proper testing methods (Gu, 2021) while evaluating the biodegradability of plastics. It is important to know that the biodegradability is highly dependent on the basic chemistry of polymers including purity, chemical composition, molecular weight and molecular weight distribution, which is not given the due importance (Ahmed et al., 2018). Additionally, the active microorganism such as fungi and bacteria (Gómez et al., 2013) and their metabolic capability is an important factor for assessment of the biodegradability of polymers because it alters the plastic chemistry including degree of polymerization and molecular weight distribution. Further, to isolate the potential microorganisms, it is necessary to conduct an initial incubation of enrichment and transfer with pure plastic as only source of carbon and energy for selection of microorganisms for eliminating non-degrading microorganisms and ultimately enriching population of capable member. Once the capable microbe is obtained, the microbiology and biochemical reactions involved can be further explained and elucidated. Therefore, any claim on biodegradation of polymers must be supported by active microorganisms together with their metabolic capabilities (Gu, 2021). The selection of proper testing method is also critical to evaluate the biodegradability of polymers due to a fact that different bioplastics have varying physico-chemical properties leading to their persistence and/or different degradation behaviour under various degradation environments and environmental conditions for testing (Orhan et al., 2000). The biodegradation testing should be based on mineralization of biopolymers and mass balance & stoichiometry analysis of bioplastic carbon (Gu, 2021). Therefore, the researchers and scientists must pay attention to the basic characterization of biopolymers, the microbiology & biochemical reactions and proper testing techniques and methods before conducting research on choice of plastics/materials to assess the biodegradation of materials.

The rate of biodegradability of bioplastics is evaluated by estimation of so-called biodegradability indexes where these biodegradability indexes are related to composition and characteristics of bioplastics such as surface morphology, molecular weight and molecular weight distribution and microorganism's activity. This is estimated by evolution of CO_2, O_2 and/or CH_4, which shows the main indexes of aerobic and anaerobic biological processes, respectively (Folino et al., 2023). Generally, conversion of carbon present in bioplastics to CO_2 and/or CH_4 is widely used test method to evaluate the biodegradability under anaerobic conditions (Ruggero et al., 2019), the O_2 consumption or CO_2 production are biodegradability indexes for aerobic environment (Tchobanoglous et al., 1993) and biochemical methane potential is widely used method to stimulate anaerobic conditions at the lab scale experiments (Folino et al., 2020a). Other methods of evaluating the biodegradability reported in the previous literature are decrease in total carbon of bioplastics (Adhikari et al., 2016), visual assessment of surface erosion or discoloration (Tosin et al., 2012) assessment of oxo-degradable products (Harrison et al., 2018), and spectroscopic spectra (Ruggero et al., 2019). Weight loss is also considered as biodegradability indicator/biodegradability index despite a fact that mass loss can take place due to abiotic process without involvement of microorganisms (Folino et al., 2020b). Clear zone formation (also known as zone of clearance) method is also used to assess the biodegradability of bioplastics which measures microbial ability to hydrolyse a specific biopolymer or assesses the biodegradation potential of different microorganisms towards biopolymer (Lucas et al., 2008). This test method is widely used to define the presence of microorganisms – degrading bioplastics, when microorganisms are isolated from specific environmental matrix and the best species available to degrade biopolymer (Penkhrue et al., 2015; Urbanek et al., 2017).

Bioplastics degrade under aerobic and anaerobic conditions where aerobic conditions produce biomass, water, and CO_2, whereas anaerobic conditions form biomass, water, CO_2, and methane (CH_4) (Tokiwa et al., 2009; Youssef et al., 2018). The time frame required for bioplastic degradation is unclear and should start by understanding the mechanism of degradation (Takkalkar, 2019). Biochemical degradation pathway is complex therefore, it is suggested that the bulk polymeric materials containing different components should be assessed systematically to evaluate the effect of biotic and abiotic factors on biodegradation and determine the chemical constituents of polymeric materials supporting microbial growth (Nampoothiri et al., 2010; Le Corre et al., 2010). The biodegradation (biochemical degradation) mechanism of plastics involves several steps, as illustrated in Fig. 1. According to Shen et al. (2010), the first step to achieve the biodegradability of plastics is breaking down of polymer chains by microorganisms to decrease their molecular weight where microorganisms defecate the extracellular enzymes depolymerizing the external cell walls of the polymers. After that, the plastic materials get degraded by aerobic/anaerobic degradation (Xiao et al., 2012). It is recommended that the degradation process takes place in three stages: biofilm formation (RB), depolymerization (RD), and mineralization (RM) or bio-assimilation. The rate of RB is monitored via physical methods, the rate of RD is measured in situ, and respirometry methods assess the rate of RM under controlled laboratory conditions. An initial biodegradation process is RB (also known as biodeterioration), a complex association produced from surface-embedded microbial cells surrounded by a self-made extracellular shell of polymeric substances, i.e., proteins, polysaccharide, and integrated inorganic or organic particles (Auras et al., 2011). A lagging phase, ranging from days to weeks, is mostly monitored before the steady state biodegradation is completed because it requires extensive duration to form biofilms and adjust its microbial population (Auras et al., 2004). The second step of biodegradation of bioplastics is hydrolytic depolymerization (enzyme-catalysed) which occurs when extracellular depolymerase catalyses the cleavage of hydrolytic bond of polymers forming monomers, dimers, and oligomers. Third step of biodegradation of bioplastics is mineralization or bio-assimilation, from where small molecules uptake is ensured by cells during bio-assimilation for reproduction, growth, and mineralization, which is collected into a single parameter (Takkalkar, 2019). The biochemical mechanism of degradation (chemical constituents) of polymeric materials is explained in detail in the previous literature (Oksman et al., 2006).

Several factors influence the rate of degradation such as properties of polymers including surface morphology, shape, crystallinity, and side-chain length of polymers as well as the characteristics of degradation environment including UV exposure, temperature, mechanical forces, nutrient levels, types of bacteria, oxygen level and pH (Tikapunya et al., 2017; Ingrao et al., 2017; Dilkes-Hoffman et al., 2019a). The effect of different factors on biodegradation of bioplastics is illustrated in detail in Table 1. The biodegradation rate of bioplastics is highly dependent of degradation environment where degradation environment can be natural and controlled where the controlled biodegradation process is simple, and the standard timescale of biodegradation can be achieved and is widely used (Harrison et al., 2018). On the other hand, the natural environment of biodegradation are considered complex processes because many factors contribute to the degradation environment where biodegradation takes place such as the landfill site or a location within a landfill, freshwater or salt water, and topsoil or deep soil (Kjeldsen et al., 2018). The rate of biodegradability for different bio-based plastics in different environments including soil and composting, is illustrated in detail in Table 2. The degradation of biodegradable plastics is generally monitored by different methods, i.e., visual inspection, respirometry systems e.g., biological oxygen demand (BOD) or CO_2 evolution, physical procedures such as mass loss, biomass growth, and soil analysis (Takkalkar, 2019; Ahmed et al., 2018). The rate of biodegradation of biopolymers is shown in Fig. 2.

4. International and national standards for different types of bioplastics

The major international standardizing bodies are the American Society for Testing and Materials (ASTM), The International Organization for Standards (ISO), and the European Committee for Standardization (CEN), which create standards. Several national standardization bodies have suggested standards integrating additional procedures for testing to operate regulations better. These national standard bodies include the Japan Bioplastic Association (JBPA), the German Institute for Standardization (DIN), and the Australian Standards (AS) (Bhagwat et al., 2020). These standards introduce benchmarks for the requirements of desirable quality products and to ensure that fraudulent market behaviour is not allowed. This review article focuses on bioplastics, biodegradable plastics, and compostable plastics, and in this section, we highlight the standards related to these three types of bioplastics.

Bioplastics are attaining high demand from the industry and great interest from research and academic communities because of their 100% biodegradability as compared to conventional polymers. Both aerobic

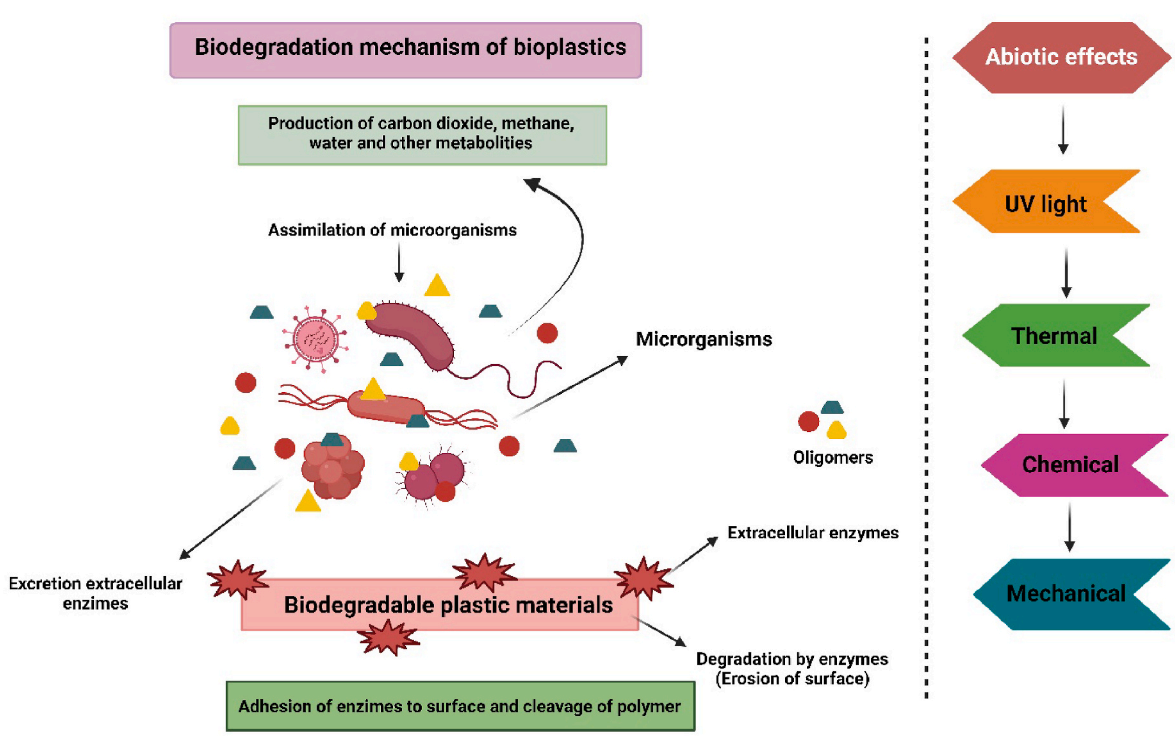

Fig. 1. Illustration for possible biodegradation mechanism of plastics (Souza, 2015).

Table 1
Summary of parameters considered for biodegradation of bioplastics.

Location		Test Method		Shape		Temp	Molecular mass	Mechanical changes	Length of study (days)	References
Lab	Field	CO_2 Evolution	Weight loss	Film	Thickness	(°C)	Changes considered or Not	Considered or Not		
✓	X	✓	X	X	X	Ambient	X	X	195	Wang et al. (2018)
X	✓	X	✓	✓	100 μm	27–30	✓	X	160	Volova et al. (2010)
✓	X	X	✓	✓	50–200 μm	25	X	✓	35	Tsuji et al. (2003)
✓	X	X	✓	✓	50 μm	25	✓	✓	300	Tsuji et al. (2002a)
X	✓	X	✓	✓	50 μm	19–26	✓	✓	150	Tsuji et al. (2002b)
✓	✓	✓	X	✓	190 μm	Field	X	X	49–100	Thellen et al. (2008)
X	✓	X	✓	✓	X	32	X	X	56	Sridewi et al. (2006)
✓	X	✓ (BOD)	X	✓	20 μm	27	✓	X	28	Sashiwa et al. (2018)
X	✓	X	✓	✓	X	17–20	✓	✓	42	Rutkowska et al. (2008)
✓	X	X	✓	✓	X	37	X	X	X	Muhamad et al. (2006)
X	✓	X	✓	X	X	6–14	✓	✓	180	Mergaert et al. (1995)
✓	X	✓	✓	✓	X	28	X	X	86	Ho et al. (2002)
✓	X	✓	X	✓	X	30	X	X	365	Greene (2012)
X	✓	X	✓	✓	50–150 μm	13–26	✓	✓	365	Doi et al. (1992)
✓	✓	✓	✓	✓	200 μm	11–20	✓	✓	180–600	Deroiné et al. (2015)
✓	✓	X	✓	X	X	4, 13, 25, 40	✓	✓	360	Deroiné et al. (2014)

Table 2
The degradation rate (biodegradability%) of biodegradable plastics in soil and compost environments.

Type of Environment	Plastic-type	Degradation conditions	Biodegradability (%)	Duration (days required to degrade)	Reference
Soil	PBS/starch powder	25°C and 60% humidity	24.4	28	Adhikari et al. (2016)
	PBS/starch films	25°C and 60% humidity	7	28	
	PBS powder	25°C and 60% humidity	16.8	28	
	PBS films	25°C and 60% humidity	1	28	
	Starch-based bioplastics	20°C and 60% moisture	14.2	110	Gómez et al. (2013)
	PHA	20°C and 60% moisture	48.5	280	
	PHA	35°C	35	60	Wu (2014)
	PHB	N/A	98	300	Boyandin et al. (2013)
	PLA	30% moisture	10	98	Wu (2012)
	PLA powder	25°C and 60% humidity	13.8	28	Adhikari et al. (2016)
	PLA/SF (90/10%)	30% humidity	>60	98	Harmaen et al. (2015)
	PLA/NPK (62.5/37.5%)	30°C and 80% humidity	37.4	56	Wu (2012)
	PLA/NPK/EFB (25/37.5/37.5%)	30°C and 80% humidity	43	56	Harmaen et al. (2015)
	PLA/rice husk biocomposite (60/40%)	35°C	>90	60	Wu (2014)
Composted soil	Nylon 4 (Polyamides, bio-based)	25°C and 80% humidity, 7.5–7.6 pH	100	120	Hashimoto et al. (2002)
Soil/Compost (90/10%)	PHA	25°C and 65% humidity	40	15	Arcos-Hernandez et al. (2012)
Compost	PLA	58°C	13	60	Ahn et al. (2011)
		58°C, 8.5 pH and 63% humidity	84	58	Kale et al. (2007)
		55°C and 70% moisture	70	28	Tabasi et al. (2015)
		58°C and 60% humidity, aerobic	60	30	Mihai et al. (2014)
	PLA/SW (70/30%), SW = Softwood	58°C and 60% humidity, aerobic	40	30	
	PHB	58°C	79.9	110	Weng et al. (2011)
		55°C and 70% moisture	80	28	Tabasi et al. (2015)
	Starch-based bioplastic made from potato almidon	58°C, aerobic	85	90	Sarasa et al. (2009)
	Starch/Resin (60/40%) bioplastic	23°C, 55% moisture and aerobic	26.9	72	Mohee et al. (2008)
	PBS	58–65°C, 50–55% moisture, 7–8 pH and aerobic	90	160	Anstey et al. (2014)
	PBS/soy meal (75/25%)			100	
	PBS/canola meal (75/25%)				
	PBS/corn gluten meal (75/25%)				
	PBS/switch grass (75/25%)			170	
	PCL	55°C	38	6	Nakasaki et al. (2006)

and anaerobic fermentation achieve the biodegradability of polymers in the presence of microorganisms that secrete extracellular enzymes to fragment polymeric chains to small molecular-weight decay products, as shown in Fig. 1 (mechanism of biodegradation of polymers). Upon disposal, the bioplastics decompose in a bio-active medium by micro-organisms such as algae, bacteria, and fungi or by marine water to humus, water, and CO_2 (Moustafa et al., 2019; Gu and Gu, 2005). According to international standards, the biodegradability environment, i.

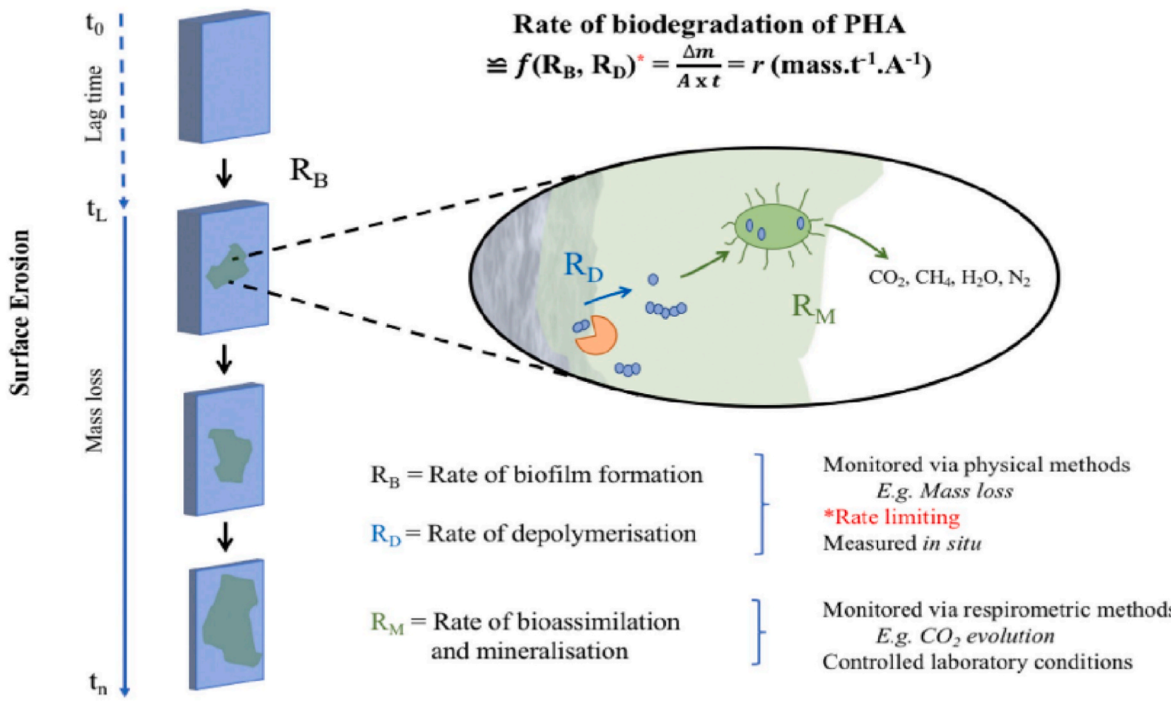

Fig. 2. Summary of degradation process for biodegradable plastics (Dilkes-Hoffman et al., 2019b), where "A" represents the surface area, "m" indicates the mass, and "t_0, t_L, t_n" are the initial time, lag time, and final time.

e., soil, compost, water, or marine environment, significantly impacts the rate of biodegradability of polymers. For instance, a higher temperature is needed for degradative reactions in compost and soil environments, increasing the degradation rate of polymers in these environments (Volova et al., 2010). The higher degradation rates in soil and compost environments may be due to high concentrations and high diversity of microbial communities in the compost and soil (Sekiguchi et al., 2011).

On the other hand, for aquatic systems, microorganisms significantly impact on the degradation of bioplastics. Therefore, it is important to consider all habitats including deep-sea, eulittoral, pelagic, sublittoral benthic, and supralittoral, along with stressors and microbial communities (Sekiguchi et al., 2011; Volova et al., 2017). Moustafa et el. (Moustafa et al., 2017a) performed biodegradability analysis of the PBE/PBAT composite in fresh water from the River Nile and marine water over 20 days. They found that the biodegradability rate was higher in the freshwater environment than in the marine water environment. This could be explained by the microbial effect of microorganisms and algae in freshwater, which attacked polymeric chains, causing cleavage of the polymeric chains (Moustafa et al., 2017b). Table 3 lists the standards and test procedures used for determining the degradation rate of different polymers in soil, marine and wastewater/sewage sludge environments. The most common standards used for the measurement of the biodegradability percentage of different polymers are ASTM D5988-12, ISO 17556:2012, UNI 11462:2012, and NF U52-001 for the soil environment, BS EN ISO 14851:2004, BS EN ISO 14852:2018, BS EN ISO 14853:2016, and BS ISO 13975:2012 for wastewater and sewage sludge, and ASTM D6691-09, ASTM D7473-12, ASTM D7991-15, ISO 18830:2016, and ISO 19679:2016 for marine environment (Table 3).

Several standards govern the compostability of plastics under a controlled industrial environment, including ASTM D6400, ISO 18606, ISO 17088:2012, EN13432. The composting standards confirm that atleast 90% organic matter should convert into CO_2 in six months' time due to biological actions during composting. Also, it requires that a 2 mm sieve retains no more than 10% of the residue after composting in three months, and the compost quality should remain the same by adding more plastic. This compost is later applied to the soil, degrading slowly, releasing essential nutrients for plant life improving the structure of soil by binding the soil particles together and improving the water retention characteristics of soil. It is recommended that if the plastics do not meet these standards' requirements, they will not be considered compostable (Jakubowicz, 2003). For instance, polyethylene plastic bags produced with starch additives are not certified as compostable plastics because they do not meet the standards of ASTM D6400. A summary of different standards used for compostable plastics under different composting conditions is given in detail in Table (S-1).

Bioplastics are bio-based and biodegradable polymers produced from renewable sources of plant biomass, products of microbially fermented and animal derived polymers emulating the biomass life cycle by producing water and CO_2. Several standards have been established by different international or national standardization bodies to define bioplastics and their degradability or compostability. For instance, the Association of the European Bioplastics describes two method to evaluate products: (i) by using test method that includes procedures which are followed by pre-defined methodology and (ii) using definitive fail/pass criteria as a requirement which needs to be met for materials/products to comply with standard (EUPB, 2018). These two systems are frequently combined, and, in such case, use of a second method ultimately states compliance of the standard. Major bioplastic standardization bodies are ASTM, CEN, JBPA, and Standards Australia; the major certification bodies for bioplastics are shown in Figure (S-1). The standards used for bioplastics are listed and explained in Table (S-2). The biodegradability standards can be classified into two types depending upon biodegradation environment and biodegradability test methods. The biodegradation environments include industrial composting, home composting, soil, fresh water, marine water and the biodegradability test methods are aerobic biodegradation, anaerobic biodegradation, disintegration and toxicity (González et al., 2023). The overview of standards available for biodegradability and compostability of biopolymer based on biodegradation environments and biodegradability test methods are listed in Table 4.

Table 3
The biodegradability standards and the test methods for all plastic materials (Harrison et al., 2018; Briassoulis et al., 2017).

Environment	Standard/ Test Method	Inoculum	Temp (°C)	Medium	Measurement Type	Test Duration	Number of replicates	Validity criteria
Soil	ASTM D5988-12	Adapted or non-adapted soil	20–28	Natural soil, Aerobic	CO_2 evolution	Max. 6 months	Not specified	- ≥70% degradation of reference material - The measured CO_2 values are within 20% of the mean at the end of the test
	ISO 17556:2012				CO_2 evolution and BOD	Max. 6 months		- ≥60% degradation of reference material - The measured CO_2 values are within 20% of the mean at the end of the test
	UNI 11462:2012	Non-adapted soil	21–28		CO_2 evolution	Max. 3 months		- ≥60% degradation of reference material - The standard deviation of replicates <10%
	NF U52-001		Not specified		Visual evidence of degradation	Max. 6 months		- ≥70% degradation of reference material - The standard deviation of replicates <10%
Wastewater and sewage sludge	BS EN ISO 14851:2004	Sludge, compost/ soil	20–25	Aerobic, synthetic	BOD, static test conditions	Max. 6 months	Minimum 2	- >60% degradation of reference material - BOD of negative control must not exceed the specified upper limit
	BS EN ISO 14852:2018				CO_2 evolution, static test conditions		Minimum 2	- >60% degradation of reference material - CO2 of negative control must not exceed the specified upper limit
	BS EN ISO 14853:2016	Sludge, livestock faeces or other organic waste	35		CO_2 and CH_4 evolution, DIC; static test conditions	Max. 3months	2	- >70% degradation of reference material - the pH of the medium must be between 6 and 8
	BS ISO 13975:2012		35 or 55	Aerobic, direct exposure to inoculum			Minimum 3	- >70% degradation of reference material after 15 days - Degradation% must differ by less than 20% between replicates
Marine	ASTM D6691-09	Seawater or preselected strains	30	Synthetic, aerobic	CO_2 evolution, static test conditions	Maximum 3 months	N/A	- ≥70% degradation of reference material
	ASTM D7473-12	Seawater or seawater and sediment	It depends on in situ conditions	Direct contact with inoculum, aerobic	Loss of dry mass, visual evidence of degradation	Maximum 6 months	3	- Not mentioned
	ASTM D7991-15	Seawater and sediment	15–28		CO_2 evolution, static test conditions	Maximum 24 months		- ≥60% degradation of reference material
	ISO 18830:2016	Sediment or Seawater and sediment		Synthetic or natural seawater	BOD, static test conditions			- >60% degradation of reference material - BOD of negative control must not exceed the specified upper limit
	ISO 19679:2016		N/A		CO_2 evolution, static test conditions			- >60% degradation of reference material - CO2 of negative control must not exceed the specified upper limit

5. Agricultural applications of biodegradable plastics, bioplastics, and compostable plastics

Plasticulture, uses plastics for agriculture applications, and utilizes 6.7 million tons of plastics annually, accounting for around 2% of global plastics production. The application of polymers in the agriculture industry includes greenhouse cover, irrigation systems, packaging, soil mulching, silage, and trailing (Zurier et al., 2021; Briassoulis et al., 2010b; Zumstein et al., 2019). The plastics used for agriculture are difficult to recycle; therefore, they are either buried in fertile soils - which deteriorates the soil properties significantly - or are landfilled (Dar et al., 2022). These practices lead to adverse environmental effects including groundwater contamination, ecosystem disturbance for terrestrial and aquatic fauna and flora, the spread of toxic microplastics, and deterioration of soil characteristics of agricultural land such as water retention capacity and gas exchange (Zurier et al., 2021; Sander, 2019). Thus, there is a global need to develop bioplastics (either compostable or biodegradable) for agricultural applications to be considered as an environmental-friendly substitute for conventional plastics. Compostable and biodegradable bioplastics possess important physical and

Table 4
An overview of standards available for definition of biopolymers, test criteria and test method to assess the biodegradability and/or compostability of bioplastics in different environments widely used in the USA, Europe and Australia (Camenzuli et al., 2023; Masroor; Cinelli et al., 2008; Lackner et al., 2023; Lindh, 2005; Briassoulis et al., 2010a).

Type of biopolymer	Australian Standards (AS)	USA Standards (ASTM)	European Standards (ISO and EN)
Bio-based biopolymer	No standard available	ASTM D6866	EN 16640 EN 16785-1 EN 16785-2
Biodegradable biopolymer	NO AS; however, ISO 17088:2012 is used in Australia	ASTM D6400-12	ISO 17088:2012 EN 13432:2000 EN 14995:2006
Compostable biopolymer	ISO 17088:2008	ASTM D6400-04	EN 13432 ISO 17088:2008
Degradability Environment	**Australian Standards (AS)**	**USA Standards (ASTM)**	**European Standards (ISO and EN)**
Industrial compost	AS 4736-2006	ASTM D6400-21	ISO 17088:2021 EN 13432:2000
Home compost	AS 5810-2010	ASTM D6400-21[a]	EN 13432:2000[a]
Soil	NO AS; however, ISO 2352:2021 is used in Australia	No standard available	ISO 23517:2021 EN 17033:2018[b]
Fresh water	No standard available	No standard available	EN 14987:2006
Marine water	No standard available	ASTM D7081-05	No standard available
Test Method	**Australian Standards (AS)**	**USA Standards (ASTM)**	**European Standards (ISO and EN)**
Aerobic biodegradation	NO AS; however, ISO 14855–2:2018 is used in Australia	ASTM D5338-15 (2021)	ISO 14855–2:2018
Anaerobic biodegradation	NO AS; however, ISO 14853:2017 is used in Australia	No standard available	ISO 14853:2017
Disintegration	NO AS; however, ISO 16929:2021 is used in Australia	NO ASTM; however, ISO 16929:2021 is used in the USA	ISO 16929:2021
Ecotoxicity	NO AS; however, EN 13432:2000 (annex E) is used in Australia	ASTM E1676-12 (2021)	EN 13432:2000 (Annex E)

a modified conditions.

b applies to only mulching films.

chemo-mechanical characteristics which make them the best substitute for traditional plastics for agricultural applications, thus reducing waste production after their service life (Rujnić-Sokele et al., 2017). Biodegradable and bio-compostable plastic materials, which meet international standards, are left in the field where they degrade by micro-organisms present in crop residue and soil resulting in higher soil fertility. The accumulation of organic matter is formed with the degradation of these materials, which is recycled back to the field (Alshehrei, 2017).

5.1. Mulching/mulch films

Plastic mulching is among the most abundantly practiced approaches in plasticulture, covering an area of more than 128,500 km^2 worldwide (Zhou et al., 2020). Plasticulture mulching is carried out for several reasons such as crop protection, preservation of water, treatment against soilborne pathogens and weeds (Achmon et al., 2017), remediation, and improving crop yields (Dar et al., 2022). For multiple applications of mulching materials, these materials must possess certain properties throughout their service life. These characteristics are mechanical flexibility (ability to be stretched in different shapes of field plots), durability (chemical, mechanical, and resistance to solvents and water), modularity (it may have varying colour, gas permeability and thickness), light weight, non-toxicity, and affordability (it must be cheaper available in abundance). Most of degradation processes occurring during mulching are complex and less understood from chemical and microbiological perspectives (Sander, 2019). However, there has been recent progress in this research area by introducing European Standard EN 17033: Plastics – biodegradable mulch films. Some examples of bio-based and biodegradable mulching bioplastics such as starch, cellulose, PLA, PHA, polybutylene, ethylene vinyl acetate (EVA), and poly-*b*-hydroxybutyrate (Sander, 2019; Kyrikou et al., 2007; Anunciado et al., 2021).

5.2. Grow bags

Among agricultural bags mostly used as plastic in agriculture, grow and fertilizer bags are frequently used. Another type of grow bag, known as a planter bag or seedling bag, is also widely used for agricultural applications where they reduce the frequency at which plants are watered, retain moisture, and control soil temperature (Spehia et al., 2018). Among conventional polymers, polyethylene (specifically low-density polyethylene - LDPE) is widely used as grow bags in agriculture; however, it produces poisonous organic materials and environmental toxicants. Thus, PHA is considered the best contender to replace LDPE for agricultural grow bags in forming operations (Astner et al., 2019). PHA has no noxious filtrate in soil; it can remove nitrogen from water favourably and does not contaminate the nearby water bodies (Ahmed et al., 2022; Dietrich et al., 2017). In addition, PHA grow bags are friendlier to roots than conventional LDPE because PHAs do not result in root deformation. Moreover, the PHA grow bags do not require duplicate handling and bag recycling after usage.

5.3. Agricultural nets

High-density polyethylene is the most widely used polymer for commercial agricultural nets (Lu et al., 2016). The tensile strength and elongation at break are important properties that influence the net formation features. Therefore, biodegradable and bio-compostable bioplastics with similar characteristics are considered an alternative to high-density polyethylene for agricultural net formation. PHA and PHA blend possess comparable elongation at break and tensile strength as high-density polyethylene; thus, PHAs and PHA blends are the best alternatives to high-density polyethylene. Among PLA derivatives and PHA blends, poly(hydroxybutyrate) and PHA/PLA blends are the most common polymers used for agricultural net formation (Cinelli et al., 2018). Furthermore, utilization of biodegradable PHA as agricultural nets is distinguished according to its compostability because these degradable plastics containing PHA compost are directly applied to soil (Ahmed et al., 2022).

5.4. Other applications

There are several other applications for polymeric materials in agriculture industry such as agricultural packaging material for liquid storage tanks, fertilizer sacks, containers, and chemical-containing cans for insecticides or pesticides (Zhao et al., 2020; Saffian et al., 2016). The plastics can be utilized as irrigation or drainage pipes for dippers, channel lining, drainage and irrigation (Eker et al., 2014; Mostafa et al., 2014), cultivated protected films for orchards and vineyards cover, direct covering and nursery film (Martín-Closas et al., 2017; Ribeiro et al., 2015), and others such as ropes and strings, balet wine, bale wraps, pots used in nursery, silage and fumigation films (OMRANI et al., 2014; Dolci et al., 2011).

6. Environmental concerns & their solution and critical analysis

The production of plastics in large quantities poses an environmental threat as it is directly associated with the greenhouse gas footprint and highly depends upon the impacts of these plastics distributed across the product lifecycle (Dilkes-Hoffman et al., 2019a). Also, polymers are non-renewable, non-biodegradable, and non-compostable; therefore, they stay in surrounding environment for centuries. Thus, a solution to these environmental threats posed by conventional plastics is necessary such as moving towards eco-friendly, renewable (biomass/plant-based), and biodegradable & sustainable plastics (Hopewell et al., 2009). The environmental friendly biopolymers were introduced in 1980s for the first time (Ghanbarzadeh et al., 2013). However, their applications are still limited for several reasons such as functional limitations and development of sophisticated methods for large-scale production (Braun et al., 2012). Robertson et al., (Robertson, 2014) suggested to focus on developing bioplastics to address the issues related to plastic waste. Another study proposes that biodegradable plastics can play an important role in solving the waste plastics problems. In the ocean, bioplastics will help in minimizing the impact on marine life due to decreased persistence than that of conventional plastics. On land, bio-degradable plastics expand available management options including anaerobic digestion and composting (Dilkes-Hoffman et al., 2019a).

It is observed in previous studies that the bioplastics possess the eco-friendly characteristics such as production of PLA saves 2/3rd of energy required to make conventional plastics (Atiwesh et al., 2021) as well as there is no net increase in CO_2 during biodegradation of PLA (Elsawy et al., 2017). Another study found that PLA emits 70% less greenhouse gases during degradation in landfills (Iwata, 2015) whereas substitution of traditional plastic with PLA produced from corn reduced the greenhouse gases by 25% (Sabbah et al., 2017). This can be attributed to a fact that the plants - from which bioplastics are produced – absorb the same amount of CO_2 during cultivation as was released during biodegradation of bioplastics.

While encouraging research and development on bioplastics, it is important to consider the environmental issues and shortcomings associated with the usage of bioplastics and biodegradation products. For instance, bioplastic production from plants will require to repurposing of land to produce bioplastics instead of meeting the food requirements (Atiwesh et al., 2021). A previous study focusing on statistical analysis of land usage for food purposes and biofuel/bioplastic production purposes found that almost 1/4th of the agricultural land used for grains production is repurposed to bioplastic and biofuel production – this practice at higher levels may affect the fulfillment of food requirements ultimately resulting in higher food prices (Popp et al., 2014). Another study assessed and compared the environmental effects of conventional plastics, bioplastics and a hybrid plastic made up of fossil fuel and renewable source and reported that the production of bioplastics generated more pollutants in terms of pesticides and fertilizers used during cultivation together with chemicals used for processing of organic material into plastics (Walker et al., 2020). The biodegradation of bioplastics can be harmful to the environment because decomposition of bioplastics in compost environment generates methane gas which is more potent than CO_2, ultimately contributing to the global warming (Lashof et al., 1990). In addition, some of bioplastics require specific environments such as specific landfill sites under certain conditions and composters or digesters running at high temperatures (Atiwesh et al., 2021; Mohanty et al., 2000). Furthermore, it is reported that bio-PET is a potential carcinogen polymer and have pernicious toxic effects on ecosystem of earth (Sabbah et al., 2017; Zhang et al., 2020).

Therefore, it is recommended to conduct the environmental assessment by monitoring resource depletion and contamination for evaluating the environmental sustainability of bioplastics during development of biopolymers (Van den Oever, 2017). The depletion of resources can be defined as the use of land, water, fuel and energy whereas the contamination factors include greenhouse gas emission, climate change, air, water and soil contamination, release of fatal chemical such as cancer-causing agents, acidification, eutrophication and production of smog (Moshood et al., 2022). This strategy ends to ensure the development of procedures, policies and packages which are sensitive and favourable to the environment (Fouad et al., 2019; Reichert et al., 2020). However, current disparities in bioplastic policies between international, national, and sub-national platforms prove that bioplastic regulation and standardization are not on the global agenda. Therefore, it is of the utmost importance to bring all the standard and certification bodies together to establish policy and set standards to define bio-based plastics, biodegradable plastics, and compostable plastics adequately.

7. Conclusion and future prospects

This article reviews the definition and performance of bio-based, biodegradable, and compostable plastics. The international and national standards recommended by standardization and certification bodies are described for all three types of bioplastics. The agricultural applications of these plastics are determined, the environmental concerns associated with using bioplastics are reviewed, and possible solutions are suggested. There are various challenges in developing bio-based, biodegradable and compostable bioplastics such as the complexity of production process and consumer interest. Biodegradable and bio-compostable materials are associated with bioplastics, which are bio-based but not necessarily degradable or compostable. The biodegradability and compostability of bioplastics can be enhanced by blending them with other materials such as proteins, chitosan, and starch. Standardization and certification bodies have recommended test methods to differentiate between bio-based, biodegradable, and compostable plastics. However, end-of-life management is still a concern for such polymers. Future research should focus on the optimal degradation and composting of bioplastics.

For future research on synthesis and industrial applications of bioplastics, it is suggested that future studies should focus on efficient production methods, look for enhancing different characteristics of these materials depending upon their applications and explore commercial applications of these plastics while considering sustainability as the main point of interest. Additionally, global guidelines and standards should be developed for the production, usage and degradation of these plastics at the end of their life considering different factors such as raw materials used, energy consumed for synthesis of these polymers and emissions from both manufacturing and their use. Also, it is important to take care of the intermediate products (by-products) generated during synthesis of bioplastics which potentially can contaminate the recycling stream if they are not separated at the source. If bioplastic waste goes to landfilling and remains unattended, it may cause environmental concerns. Therefore, instead of following the same trend as for conventional plastics to deal with their waste, it is recommended to consider them as a feedstock for other products, such as feed material for synthesis of biofuels via gasification and pyrolysis. In addition, the long-term goal of achieving sustainability of bioplastics is completely dependent on societal acceptance; therefore, it is important to educate the rural and urban communities worldwide regarding bioplastics i.e., for the usage, composition, environmental benefits, degradation and waste sorting techniques at household/consumer levels.

CRediT authorship contribution statement

Sabzoi Nizamuddin: Writing – review & editing, Writing – original draft, Conceptualization. **Abdul Jabbar Baloch:** Writing – review & editing, Formal analysis. **Chengrong Chen:** Writing – review & editing, Supervision, Project administration. **Muhammad Arif:** Writing – review & editing, Validation. **Nabisab Mujawar Mubarak:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ibiod.2024.105887.

References

Achmon, Y., et al., 2017. Weed seed inactivation in soil mesocosms via biosolarization with mature compost and tomato processing waste amendments. Pest Manag. Sci. 73 (5), 862–873.

Adhikari, D., et al., 2016. Degradation of bioplastics in soil and their degradation effects on environmental microorganisms. J. Agric. Chem. Environ. 5 (1), 23–34.

Ahmed, T., et al., 2018. Biodegradation of plastics: current scenario and future prospects for environmental safety. Environmental science and pollution research 25, 7287–7298.

Ahmed, S.R., et al., 2022. Importance and applications of biodegradable materials and bioplastics from renewable resources. Biodegradable Materials and Their Applications 789–836.

Ahn, H., et al., 2011. Biodegradability of injection molded bioplastic pots containing polylactic acid and poultry feather fiber. Bioresour. Technol. 102 (7), 4930–4933.

Alshehrei, F., 2017. Biodegradation of synthetic and natural plastic by microorganisms. Journal of Applied & Environmental Microbiology 5 (1), 8–19.

Anstey, A., et al., 2014. Processability and biodegradability evaluation of composites from poly (butylene succinate)(PBS) bioplastic and biofuel co-products from Ontario. J. Polym. Environ. 22 (2), 209–218.

Anunciado, M.B., et al., 2021. Deterioration of soil-biodegradable mulch films during storage and its impact on specialty crop production. HortTechnology 31 (6), 798–809.

Arcos-Hernandez, M.V., et al., 2012. Biodegradation in a soil environment of activated sludge derived polyhydroxyalkanoate (PHBV). Polym. Degrad. Stabil. 97 (11), 2301–2312.

Astner, A., et al., 2019. Mechanical formation of micro-and nano-plastic materials for environmental studies in agricultural ecosystems. Sci. Total Environ. 685, 1097–1106.

Atiwesh, G., et al., 2021. Environmental impact of bioplastic use: a review. Heliyon 7 (9), e07918.

Auras, R., Harte, B., Selke, S., 2004. An overview of polylactides as packaging materials. Macromol. Biosci. 4 (9), 835–864.

Auras, R.A., et al., 2011. Poly (Lactic Acid): Synthesis, Structures, Properties, Processing, and Applications. John Wiley & Sons, pp. 1–297.

Bhagwat, G., et al., 2020. Benchmarking bioplastics: a natural step towards a sustainable future. J. Polym. Environ. 28 (12), 3055–3075.

Boyandin, A.N., et al., 2013. Microbial degradation of polyhydroxyalkanoates in tropical soils. Int. Biodeterior. Biodegrad. 83, 77–84.

Braun, B., Dorgan, J.R., Hollingsworth, L.O., 2012. Supra-molecular ecobionanocomposites based on polylactide and cellulosic nanowhiskers: synthesis and properties. Biomacromolecules 13 (7), 2013–2019.

Briassoulis, D., Dejean, C., Picuno, P., 2010a. Critical review of norms and standards for biodegradable agricultural plastics part II: composting. J. Polym. Environ. 18 (3), 364–383.

Briassoulis, D., Dejean, C., 2010b. Critical review of norms and standards for biodegradable agricultural plastics part I. Biodegradation in soil. J. Polym. Environ. 18, 384–400.

Briassoulis, D., Degli Innocenti, F., 2017. Standards for soil biodegradable plastics. In: Soil Degradable Bioplastics for a Sustainable Modern Agriculture. Springer, pp. 139–168.

Camenzuli, L., Syme, R., Downie, R., 2023. Biodegradability of Plastics: Discussion Paper, pp. 1–111.

Christensen, P.R., et al., 2019. Closed-loop recycling of plastics enabled by dynamic covalent diketoenamine bonds. Nat. Chem. 11 (5), 442–448.

Cinelli, P., et al., 2008. Biodegradable and biobased polymers: definitions, standards, and future perspectives. Bionanotechnology to Save the Environment 19, 105.

Cinelli, P., et al., 2018. Biopolyesters and bio based additives based blends and composites for application in packaging and agriculture. In: 7th World Congress on Biopolymers and Bioplastics.

Dar, S.U., et al., 2022. On the quest for novel bio-degradable plastics for agricultural field mulching. Front. Bioeng. Biotechnol. 10.

Deroiné, M., et al., 2014. Seawater accelerated ageing of poly (3-hydroxybutyrate-co-3-hydroxyvalerate). Polym. Degrad. Stabil. 105, 237–247.

Deroiné, M., et al., 2015. Natural degradation and biodegradation of poly (3-hydroxybutyrate-co-3-hydroxyvalerate) in liquid and solid marine environments. J. Polym. Environ. 23 (4), 493–505.

Dietrich, K., et al., 2017. Producing PHAs in the bioeconomy—towards a sustainable bioplastic. Sustain. Prod. Consum. 9, 58–70.

Dilkes-Hoffman, L., et al., 2019a. The role of biodegradable plastic in solving plastic solid waste accumulation. In: Plastics to Energy. Elsevier, pp. 469–505.

Dilkes-Hoffman, L.S., et al., 2019b. The rate of biodegradation of PHA bioplastics in the marine environment: a meta-study. Mar. Pollut. Bull. 142, 15–24.

Doi, Y., et al., 1992. Biodegradation of microbial polyesters in the marine environment. Polym. Degrad. Stabil. 36 (2), 173–177.

Dolci, P., et al., 2011. Microbial dynamics during aerobic exposure of corn silage stored under oxygen barrier or polyethylene films. Appl. Environ. Microbiol. 77 (21), 7499–7507.

Eker, B., Alkan, M., Eker, A.A., 2014. Developing flat drip irrigation pipes using bioplastic materials. Advances in Materials Science and Applications 3, 62–74.

Elsawy, M.A., et al., 2017. Hydrolytic degradation of polylactic acid (PLA) and its composites. Renew. Sustain. Energy Rev. 79, 1346–1352.

EUPB, 2018. In: Bioplastics – Industry Standards & Labels, in Relevant Standards and Labels for Bio-Based and Biodegradable Plastics.

Folino, A., Fazzino, F., Komilis, D., 2020a. Preliminary evaluation of the anaerobic biodegradability of three biobased materials used for the production of disposable plastics. J. Hazard Mater. 390, 121653.

Folino, A., et al., 2020b. Biodegradation of wasted bioplastics in natural and industrial environments: a review. Sustainability 12 (15), 6030.

Folino, A., Pangallo, D., Calabrò, P.S., 2023. Assessing bioplastics biodegradability by standard and research methods: current trends and open issues. J. Environ. Chem. Eng. 11 (2), 109424.

Fouad, D., Farag, M., 2019. Design for sustainability with biodegradable composites. Design and Manufacturing 39–55.

García, N., et al., 2015. Biodegradable starch nanocomposites. In: Eco-friendly Polymer Nanocomposites. Springer, pp. 17–77.

Garcia, J.M., Robertson, M.L., 2017. The future of plastics recycling. Science 358 (6365), 870–872.

Ghanbarzadeh, B., Almasi, H., 2013. *Biodegradable polymers.* Biodegradation-Life of Science, pp. 141–185.

Goel, V., et al., 2021. Biodegradable/bio-plastics: myths and realities. J. Polym. Environ. 29 (10), 3079–3104.

Gómez, E.F., Michel Jr., F.C., 2013. Biodegradability of conventional and bio-based plastics and natural fiber composites during composting, anaerobic digestion and long-term soil incubation. Polym. Degrad. Stabil. 98 (12), 2583–2591.

González, F.J., et al., 2023. Assessment of biodegradability in polymers mechanisms and analytical methods. In: Biodegradable Polymers. CRC Press, pp. 272–288.

Greene, J., 2007. Biodegradation of compostable plastics in green yard-waste compost environment. J. Polym. Environ. 15 (4), 269–273.

Greene, J., 2012. PLA and PHA biodegradation in the marine environment. Department of Resources Recycling and Recovery, Sacramento, CA, USA, pp. 1–38.

Gu, J.-D., 2017. Biodegradability of plastics: the pitfalls. Applied Environmental Biotechnology 2 (2), 59–61.

Gu, J.-D., 2021. Biodegradability of plastics: the issues, recent advances, and future perspectives. Environ. Sci. Pollut. Control Ser. 28 (2), 1278–1282.

Gu, J.-G., Gu, J.-D., 2005. Methods currently used in testing microbiological degradation and deterioration of a wide range of polymeric materials with various degree of degradability: a review. J. Polym. Environ. 13, 65–74.

Guan, Q.F., et al., 2021. Microplastics release from victuals packaging materials during daily usage. EcoMat 3 (3), e12107.

Harmaen, A.S., et al., 2015. Thermal and biodegradation properties of poly (lactic acid)/fertilizer/oil palm fibers blends biocomposites. Polym. Compos. 36 (3), 576–583.

Harrison, J.P., et al., 2018. Biodegradability standards for carrier bags and plastic films in aquatic environments: a critical review. R. Soc. Open Sci. 5 (5), 171792.

Hashimoto, K., et al., 2002. Biodegradation of nylon4 and its blend with nylon6. J. Appl. Polym. Sci. 86 (9), 2307–2311.

Hillmyer, M.A., 2017. The promise of plastics from plants. Science 358 (6365), 868–870.

Ho, Y.-H., Gan, S.-N., Tan, I.K., 2002. Biodegradation of a medium-chain-length polyhydroxyalkanoate in tropical river water. Applied biochemistry and biotechnology 102 (1), 337–347.

Hopewell, J., Dvorak, R., Kosior, E., 2009. Plastics recycling: challenges and opportunities. Phil. Trans. Biol. Sci. 364 (1526), 2115–2126.

Ingrao, C., Gigli, M., Siracusa, V., 2017. An attributional Life Cycle Assessment application experience to highlight environmental hotspots in the production of

foamy polylactic acid trays for fresh-food packaging usage. J. Clean. Prod. 150, 93–103.
Iwata, T., 2015. Biodegradable and bio-based polymers: future prospects of eco-friendly plastics. Angew. Chem. 54 (11), 3210–3215.
Jakubowicz, I., 2003. Evaluation of degradability of biodegradable polyethylene (PE). Polym. Degrad. Stabil. 80 (1), 39–43.
Kale, G., et al., 2007. Biodegradability of polylactide bottles in real and simulated composting conditions. Polym. Test. 26 (8), 1049–1061.
Kjeldsen, A., et al., 2018. A review of standards for biodegradable plastics. Ind. Biotechnol. Innov. Cent 33 (1).
Kyrikou, I., Briassoulis, D., 2007. Biodegradation of agricultural plastic films: a critical review. J. Polym. Environ. 15, 125–150.
Lackner, M., Mukherjee, A., Koller, M., 2023. What are "bioplastics"? Defining renewability, biosynthesis, biodegradability, and biocompatibility. Polymers 15 (24), 4695.
Lashof, D.A., Ahuja, D.R., 1990. Relative contributions of greenhouse gas emissions to global warming. Nature 344 (6266), 529–531.
Law, K.L., 2017. Plastics in the marine environment. Ann. Rev. Mar. Sci 9, 205–229.
Le Corre, D., Bras, J., Dufresne, A., 2010. Starch nanoparticles: a review. Biomacromolecules 11 (5), 1139–1153.
Lindh, H., 2005. Evaluation of the Biopolymer Poly-Hydroxi-Alkanoates, pp. 1–66.
Lu, X., et al., 2016. Morphology and properties of bio-based poly (lactic acid)/high-density polyethylene blends and their glass fiber reinforced composites. Polym. Test. 54, 90–97.
Lucas, N., et al., 2008. Polymer biodegradation: mechanisms and estimation techniques–A review. Chemosphere 73 (4), 429–442.
MacLeod, M., et al., 2021. The global threat from plastic pollution. Science 373 (6550), 61–65.
Martín-Closas, L., Costa, J., Pelacho, A.M., 2017. Agronomic effects of biodegradable films on crop and field environment. In: Soil Degradable Bioplastics for a Sustainable Modern Agriculture. Springer, pp. 67–104.
Masroor, S., Overview of biopolymers: properties and applications, in Biopolymers in Sustainable Corrosion Inhibition. CRC Press. p. 1-13..
Mergaert, J., et al., 1995. In situ biodegradation of poly (3-hydroxybutyrate) and poly (3-hydroxybutyrate-co-3-hydroxyvalerate) in natural waters. Can. J. Microbiol. 41 (13), 154–159.
Mihai, M., Legros, N., Alemdar, A., 2014. Formulation-properties versatility of wood fiber biocomposites based on polylactide and polylactide/thermoplastic starch blends. Polym. Eng. Sci. 54 (6), 1325–1340.
Mohanty, A., Misra, M.a., Hinrichsen, G., 2000. Biofibres, biodegradable polymers and biocomposites: an overview. Macromol. Mater. Eng. 276 (1), 1–24.
Mohee, R., et al., 2008. Biodegradability of biodegradable/degradable plastic materials under aerobic and anaerobic conditions. Waste Management 28 (9), 1624–1629.
Moshood, T.D., et al., 2022. Sustainability of biodegradable plastics: new problem or solution to solve the global plastic pollution? Current Research in Green and Sustainable Chemistry 5, 100273.
Mostafa, H., Abdelrahman, H., 2014. Effect of some environmental and agricultural factors on biodegradable-drip irrigation tubes. Int. J. Agric. Sci. Res. IJASR 4, 125–136.
Moustafa, H., et al., 2017a. Facile modification of organoclay and its effect on the compatibility and properties of novel biodegradable PBE/PBAT nanocomposites. Eur. Polym. J. 87, 188–199.
Moustafa, H., Guizani, C., Dufresne, A., 2017b. Sustainable biodegradable coffee grounds filler and its effect on the hydrophobicity, mechanical and thermal properties of biodegradable PBAT composites. J. Appl. Polym. Sci. 134 (8).
Moustafa, H., et al., 2019. Eco-friendly polymer composites for green packaging: future vision and challenges. Compos. B Eng. 172, 16–25.
Muhamad, I.I., Lee, K.J., Noor, M.A.M., 2006. Comparing the degradation of poly-β-(hydroxybutyrate), poly-β–(hydroxybutyrate-co-valerate)(PHBV) and PHBV/cellulose triacetate blend. Malaysian Polymer Journal 1 (1), 39–46.
Nakasaki, K., et al., 2006. Synergy of two thermophiles enables decomposition of poly-ε-caprolactone under composting conditions. FEMS Microbiol. Ecol. 58 (3), 373–383.
Nampoothiri, K.M., Nair, N.R., John, R.P., 2010. An overview of the recent developments in polylactide (PLA) research. Bioresour. Technol. 101 (22), 8493–8501.
Narayn, R., Pettigrew, C.A., 1999. ASTM standards help define and grow a new degradable plastics industry. ASTM Stand. News 27 (12).
Nazareth, M.C., et al., 2022. Key issues for bio-based, biodegradable and compostable plastics governance. J. Environ. Manag. 322, 116074.
Oksman, K., et al., 2006. Manufacturing process of cellulose whiskers/polylactic acid nanocomposites. Compos. Sci. Technol. 66 (15), 2776–2784.
Omrani, F.H., Ghazanfari, M.A., Yosefi, A.M., 2014. Investigating the Mechanical Properties and Degradability of Bioplastics Made from Wheat Straw Cellulose and Date Palm Fiber, pp. 21–29.
Orhan, Y., Büyükgüngör, H., 2000. Enhancement of biodegradability of disposable polyethylene in controlled biological soil. Int. Biodeterior. Biodegrad. 45 (1–2), 49–55.
Penkhrue, W., et al., 2015. Isolation and screening of biopolymer-degrading microorganisms from northern Thailand. World J. Microbiol. Biotechnol. 31, 1431–1442.
Popp, J., et al., 2014. The effect of bioenergy expansion: food, energy, and environment. Renewable and sustainable energy reviews 32, 559–578.
Ragaert, K., Delva, L., Van Geem, K., 2017. Mechanical and chemical recycling of solid plastic waste. Waste management 69, 24–58.
Rahimi, A., García, J.M., 2017. Chemical recycling of waste plastics for new materials production. Nat. Rev. Chem 1 (6), 46.
Rajisha, K., et al., 2014. Preparation and characterization of potato starch nanocrystal reinforced natural rubber nanocomposites. Int. J. Biol. Macromol. 67, 147–153.
Reichert, C.L., et al., 2020. Bio-based packaging: materials, modifications, industrial applications and sustainability. Polymers 12 (7), 1558.
Ribeiro, W.X., et al., 2015. Characterization of biodegradable film based on zein and oleic acid added with nanocarbonate. Ciência Rural. 45, 1890–1894.
Robertson, G.L., 2014. Biobased but not biodegradable. Food Technol. 68 (6), 61–70.
Ruggero, F., Gori, R., Lubello, C., 2019. Methodologies to assess biodegradation of bioplastics during aerobic composting and anaerobic digestion: a review. Waste Manag. Res. 37 (10), 959–975.
Rujnić-Sokele, M., Pilipović, A., 2017. Challenges and opportunities of biodegradable plastics: a mini review. Waste Manag. Res. 35 (2), 132–140.
Rutkowska, M., et al., 2008. Environmental degradation of blends of atactic poly [(R, S)-3-hydroxybutyrate] with natural PHBV in Baltic Sea water and compost with activated sludge. J. Polym. Environ. 16 (3), 183–191.
Sabbah, M., Porta, R., 2017. Plastic pollution and the challenge of bioplastics. J. Appl. Biotechnol. Bioeng 2 (3), 00033.
Saffian, H.A., et al., 2016. Characterisation and biodegradation of poly (lactic acid) blended with oil palm biomass and fertiliser for bioplastic fertiliser composites. Bioresources 11 (1), 2055–2070.
Sander, M., 2019. Biodegradation of polymeric mulch films in agricultural soils: concepts, knowledge gaps, and future research directions. Environmental science & technology 53 (5), 2304–2315.
Sarasa, J., Gracia, J.M., Javierre, C., 2009. Study of the biodisintegration of a bioplastic material waste. Bioresour. Technol. 100 (15), 3764–3768.
Sardon, H., Dove, A.P., 2018. Plastics recycling with a difference. Science 360 (6387), 380–381.
Sashiwa, H., et al., 2018. Microbial degradation behavior in seawater of polyester blends containing poly (3-hydroxybutyrate-co-3-hydroxyhexanoate)(PHBHHx). Mar. Drugs 16 (1), 34.
Sekiguchi, T., et al., 2011. Biodegradation of aliphatic polyesters soaked in deep seawaters and isolation of poly (ε-caprolactone)-degrading bacteria. Polym. Degrad. Stabil. 96 (7), 1397–1403.
Seltenrich, N., 2016. A clearer picture of China's air: using satellite data and ground monitoring to estimate PM2. 5 over Time, 124. National Institute of Environmental Health Sciences, p. A38.
Shen, L., Haufe, J., Patel, M.K., 2009. Product overview and market projection of emerging bio-based plastics PRO-BIP 2009. Report for European polysaccharide network of excellence (EPNOE) and European bioplastics 243, 1–245.
Shen, L., Worrell, E., Patel, M., 2010. Present and future development in plastics from biomass. Biofuels, Bioproducts and Biorefining: Innovation for a sustainable economy 4 (1), 25–40.
Song, J., et al., 2009. Biodegradable and compostable alternatives to conventional plastics. Phil. Trans. Biol. Sci. 364 (1526), 2127–2139.
Souza, V., 2015. Thesis Plan Proposal—Development of a Novel Bionanocomposite Based on Chitosan/Montmorilonite with Antioxidant Activity for Food Appliances.
Spehia, R.S., et al., 2018. Lettuce growth and yield in hoagland solution with an organic concoction. Int. J. Veg. Sci. 24 (6), 557–566.
Sridewi, N., Bhubalan, K., Sudesh, K., 2006. Degradation of commercially important polyhydroxyalkanoates in tropical mangrove ecosystem. Polym. Degrad. Stabil. 91 (12), 2931–2940.
Storz, H., 2013. Bio-based plastics: status, challenges and trends. Appl. Agri. Forestry Res. 4 (63), 321–332.
Stubbins, A., et al., 2021. Plastics in the earth system. Science 373 (6550), 51–55.
Tabasi, R.Y., Ajji, A., 2015. Selective degradation of biodegradable blends in simulated laboratory composting. Polym. Degrad. Stabil. 120, 435–442.
Takkalkar, P.A., 2019. Development of Poly (Lactic Acid) and Starch Nanocrystals Based Nanocomposites. RMIT University.
Tchobanoglous, G., Theisen, H., Vigil, S., 1993. Integrated Solid Waste Management: Engineering Principles and Management Issues, pp. 1–978.
Thellen, C., et al., 2008. A processing, characterization and marine biodegradation study of melt-extruded polyhydroxyalkanoate (PHA) films. J. Polym. Environ. 16 (1), 1–11.
Tikapunya, T., et al., 2017. Molecular structures and properties of starches of Australian wild rice. Carbohydrate polymers 172, 213–222.
Tokiwa, Y., et al., 2009. Biodegradability of plastics. Int. J. Mol. Sci. 10 (9), 3722–3742.
Tosin, M., et al., 2012. Laboratory test methods to determine the degradation of plastics in marine environmental conditions. Front. Microbiol. 3, 225.
Tsuji, H., Suzuyoshi, K., 2002a. Environmental degradation of biodegradable polyesters 1. Poly (ε-caprolactone), poly [(R)-3-hydroxybutyrate], and poly (L-lactide) films in controlled static seawater. Polym. Degrad. Stabil. 75 (2), 347–355.
Tsuji, H., Suzuyoshi, K., 2002b. Environmental degradation of biodegradable polyesters 2. Poly (ε-caprolactone), poly [(R)-3-hydroxybutyrate], and poly (L-lactide) films in natural dynamic seawater. Polym. Degrad. Stabil. 75 (2), 357–365.
Tsuji, H., Suzuyoshi, K., 2003. Environmental degradation of biodegradable polyesters. IV. The effects of pores and surface hydrophilicity on the biodegradation of poly (ε-caprolactone) and poly [(R)-3-hydroxybutyrate] films in controlled seawater. J. Appl. Polym. Sci. 90 (2), 587–593.
Urbanek, A.K., et al., 2017. Isolation and characterization of Arctic microorganisms decomposing bioplastics. Amb. Express 7, 1–10.
Van den Oever, M., et al., 2017. Bio-based and biodegradable plastics: facts and figures: focus on food packaging in The Netherlands. Wageningen Food & Biobased Research 1722, 1–65.
Volova, T., et al., 2010. Biodegradation of polyhydroxyalkanoates (PHAs) in tropical coastal waters and identification of PHA-degrading bacteria. Polym. Degrad. Stabil. 95 (12), 2350–2359.

Volova, T.G., et al., 2017. Microbial degradation of polyhydroxyalkanoates with different chemical compositions and their biodegradability. Microb. Ecol. 73, 353–367.
Walker, S., Rothman, R., 2020. Life cycle assessment of bio-based and fossil-based plastic: a review. J. Clean. Prod. 261, 121158.
Wang, S., et al., 2018. Biodegradation of poly (3-hydroxybutyrate-co-3-hydroxyhexanoate) plastic under anaerobic sludge and aerobic seawater conditions: gas evolution and microbial diversity. Environmental science & technology 52 (10), 5700–5709.
Wang, C., et al., 2022. Nanocatalyzed upcycling of the plastic wastes for a circular economy. Coord. Chem. Rev. 458, 214422.
Webber, M.J., Tibbitt, M.W., 2022. Dynamic and reconfigurable materials from reversible network interactions. Nat. Rev. Mater. 1–16.
Weng, Y.-X., Wang, X.-L., Wang, Y.-Z., 2011. Biodegradation behavior of PHAs with different chemical structures under controlled composting conditions. Polym. Test. 30 (4), 372–380.
Wu, C.S., 2012. Preparation, characterization, and biodegradability of renewable resource-based composites from recycled polylactide bioplastic and sisal fibers. J. Appl. Polym. Sci. 123 (1), 347–355.
Wu, C.-S., 2014. Preparation and characterization of polyhydroxyalkanoate bioplastic-based green renewable composites from rice husk. J. Polym. Environ. 22 (3), 384–392.
Xiao, L., et al., 2012. Poly (lactic acid)-based biomaterials: synthesis, modification and applications. Biomedical science, engineering and technology 11, 247–282.
Yano, J., et al., 2014. Greenhouse gas emissions from the treatment of household plastic containers and packaging: replacement with biomass-based materials. Waste Manag. Res. 32 (4), 304–316.
Youssef, A.M., El-Sayed, S.M., 2018. Bionanocomposites materials for food packaging applications: concepts and future outlook. Carbohydrate polymers 193, 19–27.
Zhang, C., et al., 2020. A sustainable solution to plastics pollution: an eco-friendly bioplastic film production from high-salt contained Spirulina sp. residues. J. Hazard Mater. 388, 121773.
Zhao, X., Cornish, K., Vodovotz, Y., 2020. Narrowing the gap for bioplastic use in food packaging: an update. Environmental science & technology 54 (8), 4712–4732.
Zhou, B., et al., 2020. Microplastics in agricultural soils on the coastal plain of Hangzhou Bay, east China: multiple sources other than plastic mulching film. J. Hazard Mater. 388, 121814.
Zumstein, M.T., Narayan, R., Kohler, H.-P.E., McNeill, K., Sander, M., 2019. Dos and do nots when assessing the biodegradation of plastics. Environ. Sci. Technol. 53, 9967–9969.
Zurier, H.S., Goddard, J.M., 2021. Biodegradation of microplastics in food and agriculture. Curr. Opin. Food Sci. 37, 37–44.